Intelligent Traffic Congestion Predictor usin

Lead Author: T. Harshini Priya, M. Sneha, R. Vikram

Faculty Guide: Prof. P. N. Murthy, Ph.D.

Institutional Department: Department of Data Science

1. PROJECT OVERVIEW & SCOPE

Metropolitan intersection flow prediction modeling metropolitan road segments as vertices in a Spatio-Temporal Graph Convolutional Network [ST-GCN] with GRU recurrent units for 30-minute predictive horizon.

2. SPATIO-TEMPORAL ARCHITECTURE

Evaluated on 86,400 sensor loop readings across major arterial corridors. Mean Absolute Error was reduced by 14.8% compared to historical ARIMA baselines.

3. SIMULATION OUTCOMES

Model identified bottlenecks with 95.2% accuracy, predicting severe congestion events with 20-minute advance warning.